



Luis Sarmiento

Economist

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SUMMARY

I am an economist focusing on causal inference and energy modeling. As an experienced professional, I have developed substantial expertise in applied econometrics, integrated assessment models, and agent-based modeling. My research focuses on critical issues such as the energy transition, climate change, digitalization, environmental justice, and the study of policies to decrease environmental externalities in urban areas.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher MTEC Department, ETH Zurich, Zurich CH	Jan 2026 – Present
Research Economist Directorate of Economic Research, Bank of Mexico, Merida MX	Sep 2024 – Jan 2026
Environmental Economist Euro-Mediterranean Center on Climate Change, Milan IT	Sep 2022 – Sep 2024
Postdoctoral Researcher Euro-Mediterranean Center on Climate Change, Milan IT	Sep 2020 – Sep 2022
Research Associate German Institute of Economic Research, Berlin DE	Apr 2017 – Sep 2020
Research Associate Hertie School of Governance, Berlin DE	Nov 2019 – Aug 2020
Business Executive Banco del Bajío SA, León MX	Sep 2012 – Sep 2014

EDUCATION

Doctor in Economics (<i>summa cum laude</i>) Berlin Institute of Technology (TU Berlin), Berlin Germany	2017–2020
M.S. Economics and Management Humboldt University of Berlin, Berlin Germany	2014–2016
B.S. International Business University of La Salle Bajío, León Mexico	2007–2012

PUBLICATIONS

The air quality effects of Uber
Sarmiento L. and Kim Y.
Journal of Urban Economics, 153, 103846, 2026

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Occupation and temperature-related mortality in Mexico

Bressler R.D., Papp A., Sarmiento L., Shrader J., and Wilson A.

Journal of Human Resources, 2026

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Policy implications of net-zero emissions: A multi-model analysis of US energy system impacts

Bistline J. et al.

Energy and Climate Change, 6, 100191, 2025

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Residential Solar Adoption: Main Determinants and the Future of Small-Scale Photovoltaics

Reissl S., Sarmiento L., and Emmerling J.

Environmental Research - Energy, 2(3), 035002, 2025

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Health and air pollutant impacts of Net Zero CO₂ by 2050 from the Energy Modeling Forum 37

Loughlin D., Sarmiento L., Emmerling J., et al.

Energy and Climate Change, 5, 100165, 2024

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Carbon Management Technology Pathways for Reaching a US Economy-Wide Net-Zero Emissions Goal

Binsted M. et al.

Energy and Climate Change, 5, 100154, 2024

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Comparing Net Zero pathways across the Atlantic: Inter-comparison between EMF-37 and ECETF

Sarmiento L. et al.

Energy and Climate Change, 5, 100144, 2024

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Equity Implications of Net-Zero Emissions: Energy Expenditures across Income Classes

Bistline J. et al.

Energy and Climate Change, 5, 100118, 2024

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Policy Reversals in Transitional Markets: Changing Marginal Cost to Physical Order Dispatch

Gutierrez-Meave R., Rosellon J., and Sarmiento L.

Economics of Energy and Environmental Policy, 13(1), 57-82, 2024

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The Air Quality and Well-Being Effects of Low Emission Zones

Sarmiento L., Wagner N., and Zaklan A.

Journal of Public Economics, 227, 105014, 2023

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Air Pollution and Court Decisions: Evidence from Ten Million Penal Cases in India

Sarmiento L. and Nowakowski A.

Environmental and Resource Economics, 86, 605-644, 2023

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The Nonlinear Effects of Pollution on Crime: Evidence from Mexico City and New York

Sarmiento L.

Environmental Research - Health, 1(2), 021001, 2023

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Congestion, Air Quality, and Space

Tassinari F. and Sarmiento L.

Handbook of Labor, Human Resources and Population Economics, 1-18, 2023

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Air Pollution and the Productivity of High-Skill Labor: Evidence from Court Hearings

Sarmiento L.

Scandinavian Journal of Economics, 124, 301-332, 2022

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Mexico and US Power Systems Under Variations in Natural Gas Prices

Sarmiento L. et al.

Energy Policy, 156, 112378, 2021

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The Role of Energy Storage in the Uptake of Renewable Energy: A Model Comparison Approach

Giarola S. et al.

Energy Policy, 151, 112159, 2021

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The Effects of the 2013 Floods on Germany's Road Freight Traffic

Gabela J.G.F. and Sarmiento L.

Transportation Research Part D, 82, 102274, 2020

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Analysing Scenarios for Renewable Energy Integration in the Mexican Energy System

Sarmiento L. et al.

Energies, 12(17), 3270, 2019

UNDER REVIEW

Emergency Room Visits and Temperature: Evidence from Mexico

Sarmiento L., Colelli F.P., and Pavanello F.

Journal of Economic Behavior and Organization (Revise and Resubmit)

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Income Shocks, Adaptation, and Temperature-Related Mortality: Evidence from the Mexican Labor Market

Sarmiento L., Gilli M., Pavanello F., and Shayegh S.

Journal of the Association of Environmental and Resource Economists (Reject and Resubmit)

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Recycling in a Globalised Economy

Joltreau E. and Sarmiento L.

Empirical Economics (Accepted)

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When the Boundary Layer Drops: Air Quality and Healthcare Use in Mexico

Basaglia P. and Sarmiento L.

Journal of Environmental Economics and Management

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The Unintended Consequences of Low Emission Zones: Evidence from Madrid Central

Santonja A., Sarmiento L., and Wagner N.

Journal of Environmental Economics and Management (Revise and Resubmit)

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The Air Quality, Health, and Labor Supply Effects of Crop-Residue Burning

Sarmiento L. and Arellano J.

American Journal of Agricultural Economics

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Migrant Networks as Climate Insurance

Ares de Parga-Regalado G., Osuna-Gomez D., and Sarmiento L.

AEA Papers and Proceedings (Invited)

WORKING PAPERS

The Convexity of Hurricane Damages

Cohen F., Sarmiento L., and Stuka Y.

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Racial Inequalities in Regional Air Pollution Exposure in Mexico

Sarmiento L. and Gutierrez-Meave R.

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The Transnational Effect of Temperature on Remittances

Ares de Parga-Regalado G., Osuna-Gomez D., and Sarmiento L.

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Inflation Smoothing Through Natural Disaster Relief Funds

Arellano J., Llamosas I., McLiberty I., Sarmiento L., and Solorzano D.

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Temperatures and Distribution Blackouts: From Mechanisms to Health Impacts

Colelli F.P., Pavanello F., and Sarmiento L.

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Indoor Air Quality and Student Welfare: The Effect of Air Purifiers in Schools

Bonan J., Granella F., Renna S., and Sarmiento L.

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The Global Economic Burden of Air Pollution

Basaglia P., Drupp M.A., Sager L., Sarmiento L., and Schaumann F.

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Rural Development and Environmental Trade-offs in Mexico's Flagship Agroforestry Programme

Meier S. and Sarmiento L.

GRANTS AND FUNDING

- **Evaluating Non-price and Economic Instruments for Demand Evening in Mobility and Transport (EVEN-DEMAND)**, Swiss Federal Office of Energy (SFOE), Principal Investigator, CHF 220,000, 2027-2029

TEACHING

Courses Taught

- **Designing Public Policy Research (Quasi-Experimental Methods)**, ETH Zurich, Fall 2026
- **Environmental Economics and Policy**, University of Bologna, Spring 2026
- **Environmental Economics and Policy**, University of Bologna, Spring 2025
- **Environmental Economics and Policy**, University of Bologna, Winter 2024
- **Global Environmental Challenges**, Politecnico Di Milano, Spring 2024
- **Environmental Economics and Policy**, University of Bologna, Spring 2024
- **Environmental Economics and Policy**, University of Bologna, Spring 2023
- **Applied Econometrics for Energy and Env. Economics**, Berlin University of Technology, Summer 2020
- **Applied Econometrics for Energy and Env. Economics**, Berlin University of Technology, Summer 2019

Visiting Scholar

- **Comparing decarbonisation scenarios across the Atlantic**, Univ. of Tokyo, Fall 2024
- **Environmental Economics and Policy**, KDI School of Public Policy, Fall 2024
- **Environmental Economics and Policy**, KDI School of Public Policy, Spring 2023
- **Environmental Economics**, University of Verona, Spring 2023

TECHNICAL SKILLS & LANGUAGES

Software: R (RStudio), Python, Quarto, Git, GAMS, Julia

Methods: Difference-in-Differences, Regression Discontinuity, Instrumental Variables, Panel Data, Machine Learning, Agent-Based Modeling, Integrated Assessment Models

Languages: Spanish (native), English (fluent), Italian (advanced), German (intermediate)

SELECTED CONFERENCES

- **World Congress of Environmental and Resource Economists**, Lisbon PT, Summer 2026
- **Gothenburg Environmental Economics Workshop**, Gothenburg SE, Summer 2026
- **Mannheim Energy Conference**, Mannheim DE, Spring 2026
- **Essen Health Conference**, Essen DE, Spring 2026
- **First Colloquium of Regional Economists**, Merida MX, Fall 2025
- **BIS Conference on Climate and the Environment**, Bogota CO, Fall 2025
- **First Latin-American Environmental Economics Conference**, Bogota CO, Fall 2025
- **Mapping Environmental and Energy Policies Workshop**, Frankfurt DE, Spring 2025
- **EAERE Annual Conference**, Leuven BE, Summer 2024
- **MEEP Workshop**, Hamburg DE, Summer 2024
- **IAERE Annual Conference**, Pescara IT, Spring 2024
- **IEA World Congress**, Medellin CO, Fall 2023
- **ICEE World Conference**, Cagliari IT, Spring 2023
- **IAERE Annual Conference**, Napoli IT, Winter 2023
- **EAERE Annual Conference**, Rimini IT, Summer 2022

REFERENCES

Christian von Hirschhausen

Professor, TU Berlin

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Moritz Drupp

Associate Professor, ETH Zurich

mdrupp@ethz.ch